

Chapter 135 — Minimally Invasive Urinary Diversion

- The share of radical cystectomies performed by a **minimally invasive approach** rose steadily in the early 2010s (National Cancer Database, NCDB).
- **Robot-assisted radical cystectomy (RARC)** with lymph node dissection is now an **established technique**; oncological outcomes are safe and effective in both early and long-term follow-up.
- The **Quality Cystectomy Score** (Roswell Park Comprehensive Cancer Center) uses star-based criteria spanning **neoadjuvant chemotherapy consultation → adequate lymph node yield → 30-day mortality**.
- Patients undergoing minimally invasive cystectomy were **more likely to receive neoadjuvant chemotherapy** (NCDB).
- **Oncological equivalence of RARC** to open cystectomy has been validated in several randomised controlled trials (RCTs), including the landmark **RAZOR** trial — but in these trials the **diversion was performed extracorporeally**; the role of **intracorporeal urinary diversion (ICUD)** was not tested by them.
- **Laparoscopic urinary diversion** dates from 1992; adoption was hindered by **prolonged operative time, limited instrument manoeuvrability and a steep learning curve** → **hybrid procedures**: laparoscopic extirpation and lymph node dissection, then diversion through a modified open approach.
- Perceived advantages of the robot-assisted approach: **less pain, minimal blood loss, earlier return of bowel function** → quicker return to previous quality of life (QoL).

Concept — *Extirpation went robotic first and was proven equivalent; diversion lagged because it is the technically hardest, longest part, which is why the chapter's question is whether the reconstruction can also be done inside.*

DECLINE IN CONTINENT RESERVOIRS

- **Continent reservoirs have declined steadily since the early 2000s**, most visibly in Europe and North America — a trend **well under way before the robotic era**.
- Robot-assisted radical prostatectomy was adopted **rapidly**; RARC was adopted **cautiously** — cystectomy patients are complex, the operation is long, and it needs an advanced facility.
- Early robotic adopters concentrated on **extirpation and lymph node removal**; diversion was neglected because of its **technical complexity, learning curve and operative time**.
- **International Robotic Cystectomy Consortium (IRCC)**: intracorporeal neobladders increased significantly over time and were performed more often with **higher annual institutional RARC volume** and in the **more recent era** (2016–2020 vs 2011–2015) — comfort evolves from extirpation to ICUD.
- Landmark RCT of open versus complete intracorporeal cystectomy (**iROC**): only about **one in nine (11%) diversions were continent**, in both arms.
- Core problem: **no high-quality evidence that continent diversion gives better QoL** than incontinent diversion; **body image has not proved critical**, and stoma appliances and education have been streamlined.

Concept — *The fall in continent diversion is not a robotic artefact: it predates robotics and is sustained by the absence of QoL evidence in favour of continent reservoirs.*

INCORPORATION OF ROBOT-ASSISTED APPROACH

- NCDB 2010–2013: **slower adoption by academic institutions**, attributed to **robot availability** (cystectomy occupies the robot far longer than kidney or prostate surgery) and many surgeons with **limited minimally invasive training**.
- Cystectomy volumes are **low**: modest median hospital volume, with individual surgeons averaging only a couple of cases per year.
- Despite slow adoption, **conversion to open surgery was rare**.
- Two readings: experience with robotic prostatectomy **paved the way** by showing how to reduce cystectomy morbidity; the downside is **suboptimal oncological surgery during the learning curve** and diversions below open-surgery standards.
- **Simplified operative steps** are thought to produce a **smaller-radius pouch** → geometrically **low-capacity, high-pressure reservoir**.
- **IRCC**: a multinational quality-assurance consortium. Early data: only a **minority of RARCs were completely intracorporeal**. Later: among robotic cystectomy-focused centres, ICUD rose from a small minority (2005) to **nearly all cases (2016)**, an **annual increase of about 11%** — driven mainly by **ileal conduit**, and starting **earlier in Europe** than North America.
- **Unresolved**: whether the robotic approach inclines surgeons toward ileal conduit at the expense of continent diversion. A high-volume single centre found **surgeon preference** explained incontinent diversion in only a **small minority** of patients eligible for continent diversion.

KEY POINTS

- Advantages of the robot-assisted approach: **less pain, minimal blood loss, earlier return of bowel function**.
- Incorporation of ICUD has increased recently — **annual increase of about 11%**.
- **Ileal conduit is the commonest diversion** performed robotically.

Concept — *Adoption tracks volume and experience: ICUD grew first where robotic cystectomy was concentrated, and first for the simplest diversion, the ileal conduit.*

PATIENT SELECTION

- Selection has **broadened as the learning curve flattened**: high body mass index (BMI), locally advanced disease, older age and prior abdominal surgery are no longer exclusions.
- Contraindication to the minimally invasive approach: **reduced cardiac and pulmonary compliance** → inability to tolerate **steep Trendelenburg**, especially with the prolonged operative time of ICUD. Mitigated by the **da Vinci Xi** system and **AirSeal**, which permit a less steep position.
- **Neobladder contraindications are the same as for open surgery**.

CONTRAINDICATIONS BY LEVEL OF DECISION

Decision	Contraindication
Minimally invasive approach	Reduced cardiopulmonary compliance → intolerance of steep Trendelenburg (relative; eased by da Vinci Xi and AirSeal)
Use of ileum	Short bowel; severe inflammatory bowel disease; post-radiation enteritis
Neobladder — absolute	Urethral involvement distal to the prostate; serum creatinine >2 mg/dL; poor hepatic function

Decision	Contraindication
Neobladder — relative	As for open surgery; not enumerated in the chapter

- The table separates three distinct decisions — **route**, **bowel segment** and **reservoir type**; the trap is treating a contraindication to one as a contraindication to all.

SURGEON CHOICE

- **Pasadena Consensus Panel** best practice: focus on **oncological outcomes first**, then add reconstruction stepwise.
- Sequence: **RARC with extracorporeal diversion → intracorporeal ileal conduit (experience of about 30–100 cases) → continent diversion (about 25–50 cases)**.
- Once proficient, acceptable overall operative time: **≤4 hours with ileal conduit; ≤5 hours with neobladder**.

Concept — *The learning pathway is oncology before reconstruction, conduit before reservoir.*

PREOPERATIVE CARE

Enhanced Recovery After Surgery

- **Enhanced recovery after surgery (ERAS)** pathways now have **more than 20 components**, tailored to cystectomy.
- Recent modifications: **early alvimopan, intraoperative goal-directed fluid therapy, immediate oral diet, reduced opioid use**.
- ERAS versus traditional pathway: **less daily opioid, less ileus, shorter stay — but more pain**.
- Breakthrough analgesia shifted from **opioid patient-controlled analgesia (PCA) and epidurals** to **high-dose paracetamol and/or ketorolac**.
- RARC with ICUD on ERAS: **shorter stay, more likely to receive a neobladder, fewer gastrointestinal complications and fewer 90-day readmissions**. ERAS participation and **younger age** independently predicted a stay of **<10 days**.
- Meta-analysis: **ERAS reduces complications and length of stay without changing overall readmission**; integrated multidisciplinary pathways improve outcomes and satisfaction.
- Caution: a stay of **<5 days** carried about a **twofold risk of 90-day major complications** — shortening stay is not an end in itself.
- Post-discharge emphasis: **trained nurse visits**, and home attention to **hydration, exercise and ambulation**.

PERIOPERATIVE BUNDLE

Element	Recommendation	Reason
Fasting	Solids until 6 h, clear fluids until 2 h before anaesthesia	—
Bowel preparation	No mechanical bowel preparation, no oral antibiotics	Electrolyte imbalance (elderly); liquid spillage when bowel opened intracorporeally; no clear benefit
Mechanical thromboprophylaxis	Intermittent pneumatic compression and stockings	—
Pharmacological thromboprophylaxis	Low-molecular-weight heparin, weight-based, before and up to 4 weeks after surgery	Avoids significant thromboembolism

Element	Recommendation	Reason
Antibiotics	Broad-spectrum intravenous, within 1 h of the procedure	—

KEY POINTS

- **Mechanical bowel preparation and oral antibiotics should be avoided.**
- **Anticoagulation** helps avoid significant thromboembolic complications.
- Preoperative broad-spectrum **intravenous antibiotics within 1 hour** of the procedure.
- **ERAS enrolment reduces length of stay, complications and readmissions.**

Concept — *ERAS works by removing interventions that delay gut recovery (bowel preparation, opioids, fasting); its limit is that discharge too early shifts complications into the community.*

PATIENT POSITION AND PORT PLACEMENT

- Set-up: **general endotracheal anaesthesia → nasogastric tube and Foley catheter → lithotomy**, arms adducted and padded, legs abducted and slightly lowered on spreader bars or stirrups → traditionally **steep Trendelenburg**.
- **da Vinci Xi** integrates table motion, allowing repositioning without undocking → a **less steep or even flat** position.
- Insufflation by **Veress needle or Hasson technique** (Hasson when there has been previous abdominal surgery).
- **Six- to seven-port transperitoneal** approach; all ports after the camera placed under vision, in a **more cephalad** orientation — aids **small-bowel manoeuvring during diversion** and **extended lymph node dissection to the aortic bifurcation**.

PORT PLACEMENT FOR INTRACORPOREAL URINARY DIVERSION (FIG. 135.1)

Port	Size	Position and use
Camera	12 mm (8 mm with Xi)	Above umbilicus, midline or left
Right and left robotic arms	8 mm each	Level of umbilicus, lateral to rectus sheath, symmetrical
Fourth robotic arm / hybrid	8 mm; 12 mm hybrid	Above and medial to anterior superior iliac spine; used alternately with an assistant instrument
Additional left port (neobladder)	12–15 mm	Above and medial to left 8-mm port; can hold a robotic arm inside the 15-mm port for stapling
Assistant	5 mm (12 mm for neobladder); second assistant 5 or 15 mm	Between camera and right robotic arm
Left upper quadrant	12 mm	Assistant
Suprapubic (short)	12 mm (stapler) or 5 mm (stents)	Bowel reanastomosis in marionette conduit and W neobladder; aligns bowel; extended to Pfannenstiel for specimen

- **"Technique of spaces"**: stepwise extirpation that aids teaching and focus. Four male spaces: **periureteric, lateral pelvic, anterior rectal, retropubic**.
- **Extended lymph node dissection to the aortic bifurcation** also eases **crossing the left ureter** to the right.
- Specimens bagged → attention to ICUD.

- **Zero-degree lens throughout**; occasionally a **30-degree down lens** for a **deep female pelvis** during extended lymph node and vascular pedicle dissection.

KEY POINTS

- A **six- to seven-port transperitoneal** approach in a **more cephalad** position helps small-bowel manoeuvring.
- A robotic arm can alternate with an assistant instrument by inserting it inside a **15-mm laparoscopic port**.
- A **zero-degree lens** is used throughout.
- Robotic arms can be **undocked** and steep Trendelenburg reduced for the **urethra-neobladder anastomosis**.

Concept — *Port geometry is set for the diversion, not the cystectomy: cephalad placement buys reach over the small bowel and up to the aortic bifurcation.*

ANATOMIC CONSIDERATIONS IN MALES

- Nerve sparing is more feasible because of anatomical studies and the robot's **three-dimensional (3D) view, magnification and manoeuvrability** in a narrow space.
- Reserved for **anatomically safe tumour location, lower tumour burden and adequate baseline sexual function**; benefits both **sexual function and neobladder continence**.
- Four danger points, in operative sequence (Fig. 135.2): **(1) origin of nerves at the medial common iliac vessels and presacral area** (extended lymph node dissection) → **(2) high ligation of the bladder vascular pedicle** (preserves the proximal neurovascular complex) → incision of peritoneum over the rectovesical pouch → **(3) athermal seminal vesicle dissection** after the vasa ampullae, sparing **Denonvilliers fascia**, where the pelvic plexus is concentrated; middle and lower seminal vesicle may be left → **(4) proximal prostatic pedicles controlled close to the prostate**: intrafascial, traction-free, athermal.
- Complete lateral preservation (**veil of Aphrodite**) and its distal extent require **apical dissection after the bladder is dropped**.
- Computer-assisted **3D reconstruction of the inferior hypogastric plexus** has mapped the pelvic nerves serving continence and erection.
- About **one-third of the external urethral sphincter** cross-section is **laterally overlapped by the dorsal venous complex** at the apex.
- **Dorso-cavernous anastomoses** matter for cavernous haemodynamics (Doppler); anastomoses link dorsal, cavernous and bulbourethral arteries — **all except the bulbourethral are contractile** (muscular wall plus nerves).
- **Cavernous nerves converge close to the urethra at the sphincter** before penetrating the pelvic floor.
- Cadaveric study: the sphincter receives branches of the **dorsal nerve of the penis** (after it leaves the pudendal nerve), as close as **3 mm from the apex and unilateral in half** → **meticulous periurethral dissection**.

ANATOMIC CONSIDERATIONS IN FEMALES

- **Pelvic splanchnic nerves** carry adrenergic and cholinergic fibres, mediating **vaginal lubrication and genital swelling**; injury reduces vaginal and vulval blood flow.

- **Inferior hypogastric plexus:** from hypogastric and pelvic splanchnic nerves, retroperitoneal → anterolateral to rectum → lateral to cervix and vaginal fornix → **lateral vaginal wall and bladder base, where it is most vulnerable.**
- **Cavernous nerves** arise along the **lateral edge of the vagina**, run in the **vesicovaginal septum** and pass through the urogenital diaphragm.
- **Continence can be preserved despite removal of the bladder neck and cranial third of the urethra;** traction on the Foley balloon defines the bladder neck and landmarks.
- The **striated intrinsic sphincter lies mainly in the mid- to distal third** of the urethra and can be preserved oncologically safely; preserve the periurethral complex and endopelvic fascia.
- Urethral nerve branches run by the lateral vaginal wall → preserve tissue **around the proximal urethra and lateral to the bladder neck**; despite controversy, **preserve innervation for continence.**
- With nerve-sparing goals: **preserve the anterior vaginal wall; avoid transvaginal extraction** → small Pfannenstiel incision.

NERVE-SPARING CYSTECTOMY: MALE VERSUS FEMALE

	Male	Female
Point of maximal vulnerability	Medial common iliac/presacral origin; seminal vesicles and Denonvilliers fascia; prostatic pedicles; apex	Lateral vaginal wall and bladder base
Key preservation manoeuvre	Athermal, intrafascial, traction-free pedicle control; meticulous periurethral apical dissection	Preserve tissue around proximal urethra, lateral to bladder neck; keep anterior vaginal wall
Sphincter	One-third overlapped by dorsal venous complex; dorsal-penile-nerve branches within 3 mm of apex	Mainly mid- to distal third ; continence survives loss of bladder neck and cranial third

Concept — *In both sexes the same autonomic plexus serves sexual function and neobladder continence; nerve sparing is therefore a continence operation as much as a potency one.*

ILEAL CONDUIT

Transfer of Left Ureter and Selection of Bowel

- **Left ureter crosses below the sigmoid mesentery and over the great vessels** to the right.
- Identify **duplex ureters** → implant separately or together according to calibre.
- Short left ureter → **retract sigmoid with the fourth arm**; a 30-degree down lens helps free the proximal left ureter.
- Segment: **12 cm of ileum, 15–20 cm proximal to the ileocaecal valve** (the figure legend gives 12–15 cm, about 15 cm from the valve).
- Leave **adequate length at the ileocaecal end** → avoids metabolic disturbance and kinking once exteriorised.
- **Avoid a redundant segment** → residual urine → stasis, infection, absorptive and metabolic complications, loss of renal function.

The Marionette Stitch

- Sequence: **60-inch 1-0 silk on a Keith needle** through the hypogastrium → through the **distal end of the segment (future stoma)** → back out through the same site → **held by an instrument, not tied.**
- Untied → **free movement** of the segment during conduit construction.

- Compensates for **fixed ports**, brings each part of the conduit into the field, and lowers the segment into the pelvis to keep work within the **optimal range of the robotic joints** — valuable when working space is limited.

Isolation of the Bowel Segment and Creation of the Ileal Conduit

- Sequence: **fourth arm holds proximal segment at stretch** against the marionette → hook cautery incises mesenteric peritoneum (a **vascular stapler** is quicker) → two mesenteric windows → **45-mm Endo GIA** divides bowel proximally and distally → cut ends held with **0-silk to prevent malrotation** → **bowel continuity not yet restored**.
- Keep orientation and **avoid narrowing the conduit base**, which limits mesenteric blood supply.
- **Conduit washout has been abandoned**: no reduction in infection, and it causes spillage during stenting and anastomosis.

Ureteroileal Anastomosis

BRICKER VERSUS WALLACE

Bricker	Wallace
Refluxing end-to-side, each ureter separately	Ureteric ends joined as a "Y", single limb to proximal conduit
Easy; keeps renal units separate	One anastomosis to the bowel

- Sequence: two openings on either side of the proximal conduit → **athermal, cold-scissors wide buttonhole enterotomy with serosal removal** (adopted to reduce strictures) → marionette lowered → ureter implanted with **minimal traction** → after clear urine efflux, **wide spatulation**, fourth arm retracting the cut distal ureter → **left side first**.
- First suture: **conduit outside-in, then ureteric angle inside-out**; conduit bite **perpendicular to the proximal staple line** → alignment, no malrotation or back-walling.
- The operating position of the anastomosis is **not the final lie** of the conduit.
- Interrupted sutures along both sides to halfway → stent; alternatively **Van Velthoven double-armed 4-0 Vicryl** (about 5 cm) from the spatulation angle along both sides.
- Stenting after the posterior wall: **metal laparoscopic suction tube via 15-mm right assistant port** into distal conduit → advanced to the anastomosis → held by needle driver → **90-cm 8.5-French (Fr) single-J stent over guidewire** to renal pelvis → secured with **3-0 chromic** → **guidewire removed only after anchoring** (no tactile feedback; stent otherwise hard to find).
- **Distal ureter excised completely** for final histopathology; **no routine frozen section**.
- Marionette turns the conduit → right side done identically. Stent ends drain via the 15-mm port and are **not clamped to the drapes** (dislodgement).
- **Peritoneal flap** preserved during right periureteric dissection → covers and **retroperitonealises** the proximal anastomosis → less local inflammation.

Restoration of the Bowel

- **Reanastomose bowel after both ureters** → easier conduit handling; reanastomosis tailored to avoid kinking or tension.
- **Robot-sewn ileoileal anastomosis** is feasible — short sewing time, no leaks, **much cheaper than staplers**.

Pre-Stoma Preparation

- Sequence: robot left docked, instruments out → surgeons scrub, **cruciate fascial incision with four stay sutures** → arms reinserted; haemostasis checked; specimen-bag strings out via suprapubic port →

vascular clamp through stoma site grasps marionette and stent ends → Jackson-Pratt drain via right assistant port → undock → **urethral Foley left as pelvic drain** → conduit delivered through stoma, **orientation checked once pneumoperitoneum is lowered.**

KEY POINTS

- The **marionette stitch is not tied** but held by an instrument, allowing free movement.
- Preserve **adequate ileocaecal-end length** → avoids metabolic disturbance and kinking.
- **Guidewire removed only after the anchoring suture** — no tactile feedback.
- **Avoid a thermal enterotomy** at the ureteric anastomosis; **retroperitonealise** the anastomosis.

Concept — *Each conduit step replaces the missing hand: the marionette substitutes for manual retraction, and the guidewire is kept until anchoring because the stent cannot be felt.*

INTRACORPOREAL NEOBLADDER

- Several intracorporeal techniques are now in practice (Table 135.1); ports as in Fig. 135.1.
- Before neobladder construction: **undock and reduce Trendelenburg to 10–15 degrees** for the urethral anastomosis — unnecessary with **Xi** and an integrated table.
- Key difference from open surgery: **early neobladder-urethral anastomosis**, advocated by the **Karolinska** group and adopted by most.

TABLE 135.1 COMPARISON OF CONFIGURATION AND TECHNICAL NUANCES OF INTRACORPOREAL URINARY DIVERSION

Institution	Configuration	Ileum (cm)	Pouch suture	Urethral anastomosis	Ureteric suture / stent	Catheter (Fr, type)	Time (min)
Karolinska	Modified Studer / amorphous	50	3-0 V-Loc	2-0 Quill	5-0 Biosyn / single J	22 Foley	420
University of Southern California (USC)	Modified Studer with rotation / globular	65	2-0 V-Loc	3-0 Monocryl	4-0 polyglactin / double J	22 Couvelaire	361
University College London Hospital (UCLH)	Pyramid	50	3-0 V-Loc	3-0 V-Loc	4-0 polydioxanone / single J or 8-Fr feeding tube	16 urethral + 18 suprapubic	330
Clinique Saint-Augustin	Modified Y	40	NR	NR	NR / single J	20 Mercier	315
Padua	Folded U	42	Stapler	2-0 Monocryl	4-0 Monocryl / double J	22 haematuria	305
Roswell Park (RPCCC)	Modified W — Hautmann	50	3-0 V-Loc	3-0 V-Loc	4-0 polyglactin / single J	22 haematuria	357
University of Florence	Asymmetrical U (FloRIN)	50	3-0 V-Loc	4-0 Monocryl	3-0 Stratafix / single J	20 Foley	410

NR, not reported; FloRIN, Florence Robotic Intracorporeal Neobladder; RPCCC, Roswell Park Comprehensive Cancer Center.

WHERE THE TEXT AND TABLE 135.1 DIFFER

Technique	Item	Table 135.1	Text / figure
Karolinska	Urethral anastomosis	2-0 Quill	4-0 Quill (Van Velthoven)
Karolinska	Stents	Single J	Externalised suprapubically or internal double J
USC	Catheter	22-Fr Couvelaire	24-Fr haematuria

- The table's purpose is **side-by-side technique recognition: 40–65 cm** of ileum, overwhelmingly **barbed V-Loc** for the pouch, operative times **305–420 minutes**.
- Pattern: **USC uses the longest segment (65 cm)**, Clinique Saint-Augustin the **shortest (40 cm)**; Padua is the **only stapled pouch**; UCLH adds a **suprapubic catheter**.
- Trap: the institutional names map to **eponymous open designs** — Karolinska and USC are **Studer**, Roswell Park is **Hautmann W**, Padua derives from the **Vesica Ileale Padovana**.

DISTINGUISHING FEATURES OF THE SEVEN NEOBLADDERS

Technique	Distance from ileocaecal valve	Signature step	Ureteric implantation
Karolinska Studer	Up to 25 cm	Early urethral anastomosis; 10-cm afferent limb; Penrose sling	Bricker or Wallace to afferent limb (end-to-end)
USC Studer	11-cm distal tag to division	44 + 16 + 5 cm; 90-degree anticlockwise rotation	Bricker, ureters crossed, double J
UCLH Pyramid	—	Koie-derived cross-fold; stay sutures 25 cm from each limb	To short limbs; stents externalised; suprapubic tube
Saint-Augustin Y	About 20 cm	Mercier catheter with two hydrophilic wires	Refluxing , direct to limb ends; ureters not crossed
Padua	≥20 cm	Stapled 10-cm inverted-U neck	Modified split-nipple (antireflux)
FloRIN	About 20 cm	Asymmetrical U (30/20 cm); L-shaped neo-trigone	Antireflux , lateral anterior aspect
Roswell Park W	12–15 cm	Four stay sutures A–D; one trough at a time	End-to-side into chimneys

Modified Studer Neobladder: The Karolinska Institute Technique

Configuration and Setup

- Ileum mobilised for a **tension-free urethral anastomosis**; Studer principles; **U-shape** measured up to 25 cm from the ileocaecal valve and up to 40 cm proximal to the urethral anastomosis.

Neobladder-Urethral Anastomosis

- **Van Velthoven technique, 4-0 Quill**; V-Loc avoided anecdotally — **sharper, larger barbs may traumatise**.
- Difficult approximation → neobladder held between **two customised Penrose drains** rolled around its urethral end.

Detubularization of Bowel

- **Distal 40 cm** detubularised on the antimesenteric border with cold scissors or over a **24-Fr chest tube**; **10-cm proximal isoperistaltic afferent limb** left intact for the ureters (left side).

Creation of Neobladder

- **Posterior plate** (running 2-0 or 3-0 V-Loc) → **distal half of anterior wall** → remainder **left open until the end**.

Uretero-Neobladder Anastomosis

- Left ureter through a **tunnel under the sigmoid mesentery**; **Bricker** to the afferent limb, or **Wallace** (spatulated, posterior walls joined). Stents **externalised just above the symphysis** or internal **double J**.

Closure of the Neobladder

- Closed once ureters are stented and a **22-Fr Foley** is confirmed in place.

Modified Studer Neobladder: The University of Southern California Institute of Urology Technique

Configuration and Setup

- **65 cm of ileum: 44 cm pouch, 16 cm chimney, 5 cm discarded** (the discard allows ileoileal anastomosis without deep mesenteric division).
- Tag the **most dependent, mobile ileum** as the urethral anastomotic site → **11 cm distally** = division site → **33 cm proximally** tagged → further **16 cm = chimney** (afferent limb).
- 44 cm opened antimesenterically over a **24-Fr chest tube**; staple lines removed; medial edges aligned with four interrupted sutures, closed with **2-0 V-Loc**; **shortest (6-inch) barbed suture** for efficiency.

Rotation of Pouch, Neobladder-Urethral Anastomosis, and Cross-Folding of the Pouch

- Posterior plate **rotated 90 degrees anticlockwise** → marked site at the **first 11-cm point** aligned to the urethra.
- **Rocco-type stitch** to the distal posterior Denonvilliers fascia reduces anastomotic tension.
- **Double-armed Monocryl on an RB1 needle** over a **24-Fr haematuria catheter**; remaining pouch closed with running **2-0 V-Loc**.

Uretero-Neobladder Anastomosis

- Left ureter **under the sigmoid colon**; both spatulated; **double J**; **Bricker**; anterior line closed once stents drain and catheter position is checked.
- USC time-savers: **11-cm premeasured stitch**, **colour-coded stay sutures**, traction-countertraction stays, **short V-Loc**.

Pyramid Neobladder: The University College London Hospital Technique

Configuration and Setup

- Detubularised **cross-folded** pouch adapted from the **Koie technique**, designed to **simplify pouch creation**; **50 cm**, most dependent part to the urethra.

Formation of the Pyramid Neobladder

- Sequence: urethral anastomosis → dependent part detubularised → **two stay sutures 25 cm from each limb** → posterior plate **3-0 V-Loc** → anterior plate closed **up to 10 cm from the symphysis** → two lateral stay sutures midway → closure **lateral to medial in a coronal plane**.

Uretero-Neobladder Anastomosis

- Spatulated ureters to the **short limbs on their own sides** over stents; stents **externalised**; **suprapubic tube** for irrigation and safety.

Modified Y Neobladder: The Clinique Saint-Augustin Technique

Configuration and Setup

- **40 cm** isolated about 20 cm from the ileocaecal valve; dependent part arranged as a **modified Y** and brought to the urethra; antimesenteric border incised and anastomosed to the urethra.

Creation of the Neobladder

- Incised limb ends joined → posterior plate → anterior reconstruction. A **20-Fr Mercier catheter** carrying **two hydrophilic guidewires** is placed for later stenting.

Uretero-Neobladder Anastomosis

- Ureters spatulated, **not crossed**, anastomosed directly to the open limb ends in a **refluxing** fashion; **two single-J stents** passed **per urethram** over the preplaced wires.

Padua Neobladder

Configuration and Setup

- About **42 cm** of ileum, **at least 20 cm** from the ileocaecal valve; isolated with **one stapler load proximally and two distally**.
- Urethral site marked **12–14 cm** from the distal end; proximal half forms the **left base and dome**. Segment layout: 8 cm right plate + 10 cm neck + 8 + 16 cm left plate.
- Sequence: **10-cm inverted-U neck** with stay sutures at 8 and 18 cm → distal 8 cm detubularised → **stapler fired to create the neck** → remaining 24 cm detubularised → distal 8 cm folded to form the **dome** → next fold forms the **posterior plate**; **22-Fr haematuria catheter**.

Uretero-Neobladder Anastomosis

- Ureters through the posterior neobladder, spatulated; **modified split-nipple** anastomosis; anterior closure gives final shape.

FloRIN: Florence Robotic Intracorporeal Neobladder

Configuration and Setup

- **50 cm** about 20 cm from the valve; urethral anastomosis over a **20-Fr Foley** sets an **asymmetrical U (30 cm distal, 20 cm proximal)**.
- Entire segment detubularised → posterior plate reconfigured as an **L** → **neo-trigone** (wider posterior support) → neck reconstructed by suturing the anterior wall (**2–5 cm** anterior and proximal) → posterior plate folded anteriorly into two symmetrical segments. **Neck length is adjusted** so the posterior plate reaches the anterior anastomosis.

Uretero-Neobladder Anastomosis

- Spatulated ureters reimplanted by an **antireflux** technique on the **lateral side of each anterior aspect**; single-J stents secured; anterior plate closed as an **inverted V**.

Modified "W" Hautmann Neobladder: The Roswell Park Comprehensive Cancer Center Technique

Configuration and Setup

- **45–50 cm**, 12–15 cm from the ileocaecal junction, divided into **right and left troughs**, each with a descending and ascending loop and a **chimney**.
- **Four stay sutures**: **A** — most dependent point of right trough; **B** — 12–15 cm proximal, upper end of right trough; **C** — creates left trough; **D** — upper end of left trough. A and C are tagged to the **Foley tip**, which acts as a **marionette**.

Creation of the Neobladder

- Sequence: right trough first, **excluding the 5-cm chimney** → cranial enterotomy, cut with (preferably) **hot scissors**, bipolar traction and suction countertraction → incision **a few millimetres from the mesenteric border** (wider surface for anterior closure) → mesenteric edges sewn **3/0 V-Loc**, **caudal to**

cranial, fourth-arm traction, tightened every third throw → left trough likewise → troughs joined → **posterior plate**.

- **One trough at a time** → less spillage, orientation maintained.
- Release tags A and C → dependent posterior plate to posterior urethral plate (**3/0 V-Loc**) → anterior anastomosis over a **22-Fr silicone catheter** → close **only the lower half** of the anterior wall.

Uretero-Neobladder Anastomosis

- Right chimney staple line removed, opening closed (3/0 Vicryl); **end-to-side** anastomosis through an **anterior chimney enterotomy**.
- Fourth arm retracts the **Hem-o-lok-clipped** ureteric end; partial transection and spatulation for a wide anastomosis; **single-armed 4-0 Vicryl (5 cm), interrupted** — or double-armed, continuous.
- First stitch **outside-in on the ureter at the spatulation angle, inside-out on the neobladder**; fourth arm brings the ureter to the enterotomy **before tying** → tension-free.
- Stenting after the posterior wall: metal suction tube via the **15-mm assistant port** or the **22-Fr Coudé haematuria catheter**, through the open anterior wall → held by needle driver → **8.5-Fr single-J or paediatric feeding tube** over a guidewire → tube withdrawn holding the stent.
- **Single J preferred**: long enough to fix distally; **3-0 chromic** secures it to the neobladder, a second stitch to the Foley for later removal → anterior half completed → left side likewise.
- Remaining anterior wall closed **3/0 V-Loc, T-shaped (inverted T)**; Foley used for irrigation; **Jackson-Pratt or Blake** drain.

Restoration of the bowel. Short **12-mm suprapubic port** → **60-mm Endo GIA side-to-side** along the antimesenteric borders → open ends closed with Endo GIA fired horizontally via right assistant port → stay suture at the antimesenteric edge → mesenteric window closed with 3-0 silk.

Manoeuvres facilitating neobladder-urethral anastomosis. A tension-free anastomosis is critical and one of the hardest steps, especially in steep Trendelenburg, where stretch must be judged **without tactile feedback**.

MANOEUVRES FOR A TENSION-FREE NEOBLADDER-URETHRAL ANASTOMOSIS

Manoeuvre	Effect / caveat
Preserve dorsal venous complex and urethral length	Cutting the dorsal venous complex without initial suturing opens the apex → maximal urethra, no apical margin
Mobilise sigmoid out of the true pelvis	Occasional lateral stay suture until neobladder built
Reduce Trendelenburg after undocking	Xi with integrated table avoids undocking
Perineal pressure ; Foley guides needle	Borrowed from robotic prostatectomy
Barbed suture (V-Loc)	Less time and complexity
Detubularise, open and stretch bowel	Adds length
Incise ileal peritoneum, release mesenteric fat	Adds length — but exposes mesenteric vessels → bleeding, threatens neobladder blood supply
Penrose drains (Karolinska)	Gentle stretch and traction

Concept — Every intracorporeal neobladder is a variation on one problem — **getting a detubularised, low-pressure reservoir to the urethra without tension**; the techniques differ chiefly in folding geometry and in whether the ureters reflux.

Robot-Assisted Intracorporeal Continent Cutaneous Diversion

Colon Mobilization, Isolation, and Ileocolic Anastomosis

- Sequence: **right colon mobilised from mid-transverse colon to ileocaecal junction** → stay sutures at caecum and hepatic flexure → greater omentum incised; proximal transverse colon freed from duodenum and retroperitoneum → hepatic flexure off **Gerota's fascia** (hepatocolic ligament) → **white line of Toldt** → terminal ileum divided **15 cm proximal to the ileocaecal junction** → transverse colon divided **between hepatic flexure and middle colic artery** → **ileocolic continuity restored immediately** (prevents malrotation).

Creation of Colonic Pouch

- Sequence: **appendix base resected** → pouch irrigated via haematuria catheter through the caecal opening → staple line excised → **detubularised along the anterior taenia coli** → **folded vertically**, distal end to the spatulated ileocaecal valve → posterior line **2-0 Vicryl and continuous V-Loc**.

Uretero-Colonic Anastomosis

- Left ureter below the sigmoid mesentery; both spatulated, **through the posterior colonic plate**, 4-0 or 5-0 Vicryl, **double-J stents**. **24-Fr R  sch catheter as caecostomy** via right lateral port, through the **appendiceal stump**, secured with a 2-0 Vicryl purse-string. Anterior line closed as the posterior.

Tapering of the Ileum and Imbricating the Ileocecal Valve

- Sequence: Allis clamp via **2-cm V-shaped umbilical skin flap** → terminal ileum exteriorised → **14-Fr Foley** through ileum into colon → ileum returned → **stretched over catheter; excess antimesenteric ileum excised with serial staplers** → **ileocaecal valve imbricated** with interrupted 3-0 Vicryl → spatulated ileum sutured to **umbilical skin** as stoma (Fig. 135.11: Indiana pouch).

Concept — *The Indiana pouch separates two functions: capacity from the detubularised right colon, continence from the tapered ileum and imbricated ileocaecal valve.*

Postoperative Care

- Optimise fluid balance and avoid overhydration** — fluid shifts in an elderly population.
- No benefit from leaving a nasogastric tube** (Cochrane meta-analysis).
- Pelvic drainage preferred. **Adequate drainage and ureteric stenting** aid bowel recovery and reduce **metabolic acidosis**; **stentless** ureteroileal anastomosis for conduits is feasible.
- Paediatric feeding tubes tied to the urethral catheter avoid cystoscopic removal; drawback — **stents lost if the Foley falls out**.
- Neobladder saline irrigation every 8 hours** → prevents mucus plugging and catheter blockage.

DRAINS, CATHETERS AND STENTS

Tube	Removal
Urethral Foley as pelvic drain (conduit)	Postoperative day 1
Blake drain (neobladder)	When output falls
Neobladder urethral catheter	3–4 weeks , after a leak-free pouchogram
Externalised ureteric stents	About 1 week
Internal double-J stents	Cystoscopically at 3–4 weeks
Paediatric feeding tubes on urethral catheter	With the catheter after acceptable pouchogram

KEY POINTS

- Optimise fluid balance, avoid overhydration** in this elderly population.

- **Pelvic drainage and ureteric stenting** aid bowel recovery and avoid metabolic abnormalities.
- **Irrigate the neobladder every 8 hours.**

Outcomes

- ICUD aims to **minimise bowel manipulation** and reduce pain through **minimal wound retraction and smaller incisions**.
- Enablers: **standardised RARC and extended pelvic lymph node dissection** with open-equivalent oncology, improved console skills, and **EndoWrist** articulation for suturing, far easier than conventional laparoscopy.

Operative Time

- Historically the **major limitation** of ICUD; the techniques in Table 135.1 take **305–420 minutes**.
- **Prolonged cystectomy time → more complications and perioperative mortality**, independent of stage or comorbidity → operative time is a **quality measure** for RARC.
- Determinants: demographics and comorbidity, stage, complexity, technical modification, **surgeon experience, hospital volume**.
- IRCC: operative time **fell over the decade**; the **strongest determinant was diversion type — ileal conduit about 70 minutes shorter than neobladder**.
- **Older, comorbid patients had shorter times** (surgeons mindful of anaesthetic risk); **women longer** (hysterectomy, vaginal reconstruction); **high BMI and prior surgery longer**.
- **No significant difference between intracorporeal and extracorporeal diversion** — undocking, repositioning and redocking for extracorporeal work absorbs the difference.
- **High-volume institutions faster**. IRCC proficiency threshold exceeded the earlier estimate of **22 RARCs**: surgeons with **≥66 RARCs** saved about **55 minutes** per conduit case → case for **fellowship training and dedicated cystectomy programmes**.

Learning Curve

- Intracorporeal neobladder: operative time and **conversion risk fell substantially after only 10 consecutive cases**; early and late complications both fell with experience.
- USC: a **consistent console surgeon roughly halved** operative time.
- **Mentoring** reduced time and avoided conversion; **conversion to open prolonged length of stay**.

Complications

- **Most complications are attributable to the diversion**. Minimally invasive advantages: less bowel manipulation, **fewer insensible losses**, less analgesia.
- Graded by **modified Clavien-Dindo** or **modified Memorial Sloan Kettering Cancer Center (MSKCC)** systems, usually to 3 months.
- **Two-thirds of significant complications needing reoperation occur after 3 months** (Roswell Park) → 90-day reporting underestimates the burden.
- IRCC over two decades: **overall and high-grade complication rates unchanged**; complicated patients more often had neobladders and reoperation, and more aggressive disease (higher node yield, variant histology, adjuvant therapy); **wound complications fell**.
- Meta-analysis of RCTs: **no difference in 90-day major complications**, open versus robotic.

TABLE 135.2 COMPARISON OF COMPLICATIONS OF INTRACORPOREAL URINARY DIVERSION (COLLAPSED)

Parameter	Pattern across series
Grading	Clavien-Dindo, modified MSKCC, or not reported

Parameter	Pattern across series
Early (≤ 30 days)	22–63%
Late (> 30 days)	5–81% (one neobladder arm 82%)
High-grade ($\geq$ grade 3)	Up to 37%
Commonest cause	Infectious (gastrointestinal and infectious in one series)
Diversion type	Where separated (UCLH), neobladder rates exceeded conduit rates

- The table shows **how wide and unstandardised** complication reporting is; the trap is quoting a single "complication rate" for ICUD. Infection is the constant.

Reoperation After Robot-Assisted Radical Cystectomy

- Most reoperations are **diversion-related and late**; in the authors' series about a quarter needed intervention, the **majority after 90 days**, with cumulative reoperation reaching nearly half by 5 years.
- Predictors: **neoadjuvant chemotherapy** and **any high-grade postoperative complication**.

Uretero-Enteric Strictures

- The authors' intracorporeal conduit had **higher stricture rates, rising over 5 years** → abandoned the **thermal-hook slit enterotomy** for an **athermal buttonhole enterotomy with serosal removal and retroperitonealisation**.
- **Robotic stricture revision is comparable to open**; **indocyanine green (ICG)** localises the ureter and stricture.

Fistula and Bowel Obstruction

- **Fistula is uncommon**, about half repaired robotically; **intervention for bowel obstruction rises with time**.

Abdominal Wall Related

- Causes of intervention: **parastomal and incisional hernia, adhesions, wound dehiscence**.
- **Parastomal hernia after RARC with conduit is common and cumulative** — about one-third by 3 years, median onset about 1 year; **only a third symptomatic**, and few of those operated.
- Risk factors: **longer operative time, fascial defect ≥ 30 mm, lower estimated glomerular filtration rate** (poor healing).
- **Robotic parastomal mesh hernioplasty** is feasible and durable; longer operating time early in experience.

Functional Outcomes

Continence after creation of neobladder.

- Cystectomy abolishes the **vesicourethral reflex** → reservoir disconnected from sphincter, spinal reflex inactive → **no desire to void**.
- Voiding cue: urine reaching **posterior urethral sensory mechanisms** (intact pudendal nerves). The **urethrosphincteric guarding reflex** contracts the sphincter → **daytime continence**.
- **At night, in deep sleep, the guarding reflex is inactive** → **continence depends on reservoir functional capacity**.
- Goal: **adequate detubularisation and capacity** → **compliance**, plus **preservation of the sphincter complex**.
- Incontinence factors: **age, reservoir volume and shape, functional urethral length, nerve supply**.
- Assess continence at **6–12 months**, when capacity is maximal. Keep **nocturnal urine output below functional capacity**; **night alarm** until fullness sensation is recognised.

- Up to **25% of men** may need **clean intermittent catheterisation (CIC)** for incomplete emptying (open series).
- Predictors of good continence (open): good preoperative **Eastern Cooperative Oncology Group (ECOG)** status and **retained sensation of filling**.

TABLE 135.3 COMPARISON OF CONTINENCE OUTCOMES FOR INTRACORPOREAL NEOBLADDER (COLLAPSED)

Outcome	Text	Table	Reason for difference
Daytime continence	68–100%	68–100%; one Studer series 17%	17% is complete pad-free day and night — stricter definition
Night-time continence	55–85%	48–85%	Padua series (48%) outside quoted range
Need for CIC	Up to 11%	0–11%	—
Pads/day; follow-up	—	0–1; 2–41 months	—

- **Night continence is consistently worse than day** — the table confirms the physiology above; the trap is expecting robotic series to beat open ones.

Intracorporeal Versus Extracorporeal Diversion

INTRACORPOREAL (ICUD) VERSUS EXTRACORPOREAL (ECUD) DIVERSION — IRCC

	Earlier IRCC analysis	Updated IRCC analysis (about half ICUD)
Readmission (30 and 90 days)	Lower with ICUD	Approach not associated on multivariable analysis
Transfusion	Lower with ICUD	Lower with ICUD
Gastrointestinal complications	Lower with ICUD — supports minimal bowel handling	—
High-grade complications	—	Higher with ICUD , falling over time; not significant on multivariable analysis
Operative time	—	Shorter with ICUD
90-day mortality	Lower with ICUD — selection bias and experience	—
Who received ICUD	Minority of institutions performed it	Healthier (American Society of Anesthesiologists, ASA, <3), high-volume centres, recent era (2013–2016)

- The early ICUD advantages were **confounded by selection**; once adjusted, the approach itself is neutral for high-grade complications and readmission.

Urodynamic Findings

- Theoretical advantages: **less bowel handling → better vascularity, less scarring → good capacity**; precise apical dissection, nerve sparing and urethral anastomosis.

URODYNAMICS OF INTRACORPOREAL NEOBLADDERS

Configuration	Bottom line
Y neobladder (first series)	Good capacity, low residual; day continence mostly achieved, night poor
Y neobladder (second series)	Modest capacity, high residual; open neobladder better for capacity and voiding
Bordeaux (modified Y)	Good capacity, residual about 100 mL; modest day and poor night continence
USC intracorporeal Studer	Good capacity but low flow, high residual , low complete continence

Configuration	Bottom line
FloRIN vs open Vesica Ileale Padovana	No difference , except higher compliance with FloRIN

- Pattern: capacity is generally adequate, but **emptying and night continence lag**; only FloRIN matched its open comparator.

Quality of Life

- Evidence limited by **absent preoperative baselines, short follow-up and non-validated tools**.
- Single-centre, patient-centred metrics to 24 months: physical domains **decline to 6 months**, then recover by 24 months in **all domains except sexual function**.
- Earlier RCTs of open versus robotic cystectomy used **open diversion**. iROC (landmark multicentre RCT, intracorporeal versus open): **worse QoL with open surgery at 5 weeks, greater disability at 5 and 12 weeks; no difference after 12 weeks**.

Concept — ICUD's measurable gains are **early and bowel-related** (recovery, gastrointestinal complications, short-term QoL); long-term complications, function and QoL converge with open surgery, and the late burden is dominated by the diversion itself.

Tissue Engineering

- Aim: **eliminate bowel** from urinary reconstruction; regenerative attempts since 2007 — small intestinal submucosa seeded with 3T3 fibroblasts (rat); tubular polypropylene mesh scaffolds (patency and survival to about 16 weeks).
- **Neo-Urinary Conduit Trial in Humans (NCT01087687)**: small enrolment; engineered conduit survived about **250 days** on average; **half had stenosis and stricture**.
- Lessons: material **stiff and non-collapsible**, **failed to generate cells** around the connective-tissue sleeve, **no blood supply**.

Robotic Bladder Auto Transplantation

- USC stepwise preclinical models: **porcine → pulsatile-perfused cadavers → heart-beating brain-dead human donors**.
- Technique: **robotic retrieval of a vascularised composite bladder allograft → back-table vascular reconstruction → robotic transplantation**; console time fell with experience; **ICG fluorescence** assessed graft perfusion; vascularity and mucosal integrity held for **over 12 hours**.
- Could avoid bowel and its complications; **immunosuppression is the major limitation**.

Concept — Both frontiers target the same root cause of diversion morbidity — **bowel in the urinary tract**; engineered conduits failed on **vascularity**, and transplantation trades bowel complications for **immunosuppression**.

FUTURE DIRECTION

- With RARC and lymph node dissection safely established, **ICUD is performed worldwide**.
- Incorporate ICUD **stepwise: ileal conduit → neobladder**.
- Understanding the **anatomical, pathophysiological and technical nuances** prevents long-term complications and **upper-tract failure**.
- ICUD's permanent place depends on **operative time** and on **optimised neobladder function** → less morbidity, quicker return of QoL.